Bloom Filter-Based MPSI

Weekly Progress Meeting 19 March 2026

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Daily Supervisor: Tjitske Koster

19-03-2026

Bibliography

[1] Jelle Vos, Jorrit van Assen, Tjitske Koster, Evangelia Anna Markatou, and Zekeriya Erkin. On the insecurity of bloom filter-based private set intersections. Cryptology ePrint Archive, 2024.

[2] Ou Ruan, Changwang Yan, Jing Zhou, and Chaohao Ai. A practical multiparty private set intersection protocol based on bloom filters for unbalanced scenarios. Applied Sciences, 13(24):13215, 2023.

[3] Florian Kerschbaum. Outsourced private set intersection using homomorphic encryption. In Proceedings of the 7th ACM Symposium on Information, Computer and Communications Security, pages 85–86, 2012.

[4] Falzon, Francesca, and Evangelia Anna Markatou. "Re-visiting authorized private set intersection: A new privacy-preserving variant and two protocols." *Proceedings on Privacy Enhancing Technologies* (2025).

[5] Koster, Tjitske Ollie, Francesca Falzon, and Evangelia Anna Markatou. “Bandwidth Efficient Partial Authorized PSI.” *Cryptology ePrint Archive* (2025).

[6] Lindell, Yehuda. "How to simulate it–a tutorial on the simulation proof technique." *Tutorials on the Foundations of Cryptography: Dedicated to Oded Goldreich* (2017): 277-346.

Progress

From the plan last week:

- ~~Implement APSI1, then start implementing APSI2~~ Implemented APSI2
- Do some writing

Progress

APSI 2 – ~~InterActive~~ Judge

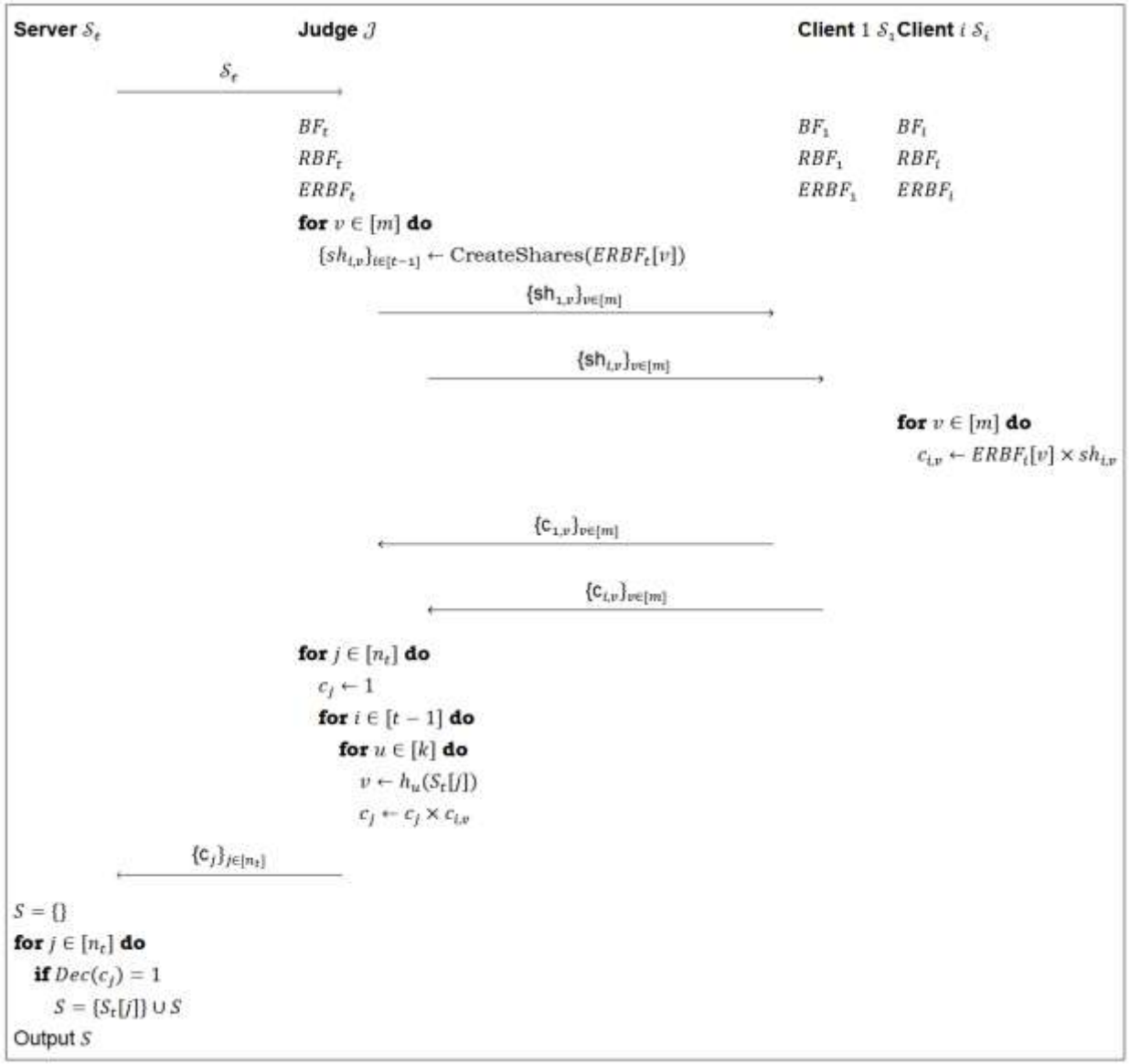
$\boldsymbol{n_t}$ = server's set size

$\boldsymbol{m}$ = bin count (BF size)

$\boldsymbol{k}$ = number of hash functions

$\boldsymbol{t}$ = number of parties (including the server)

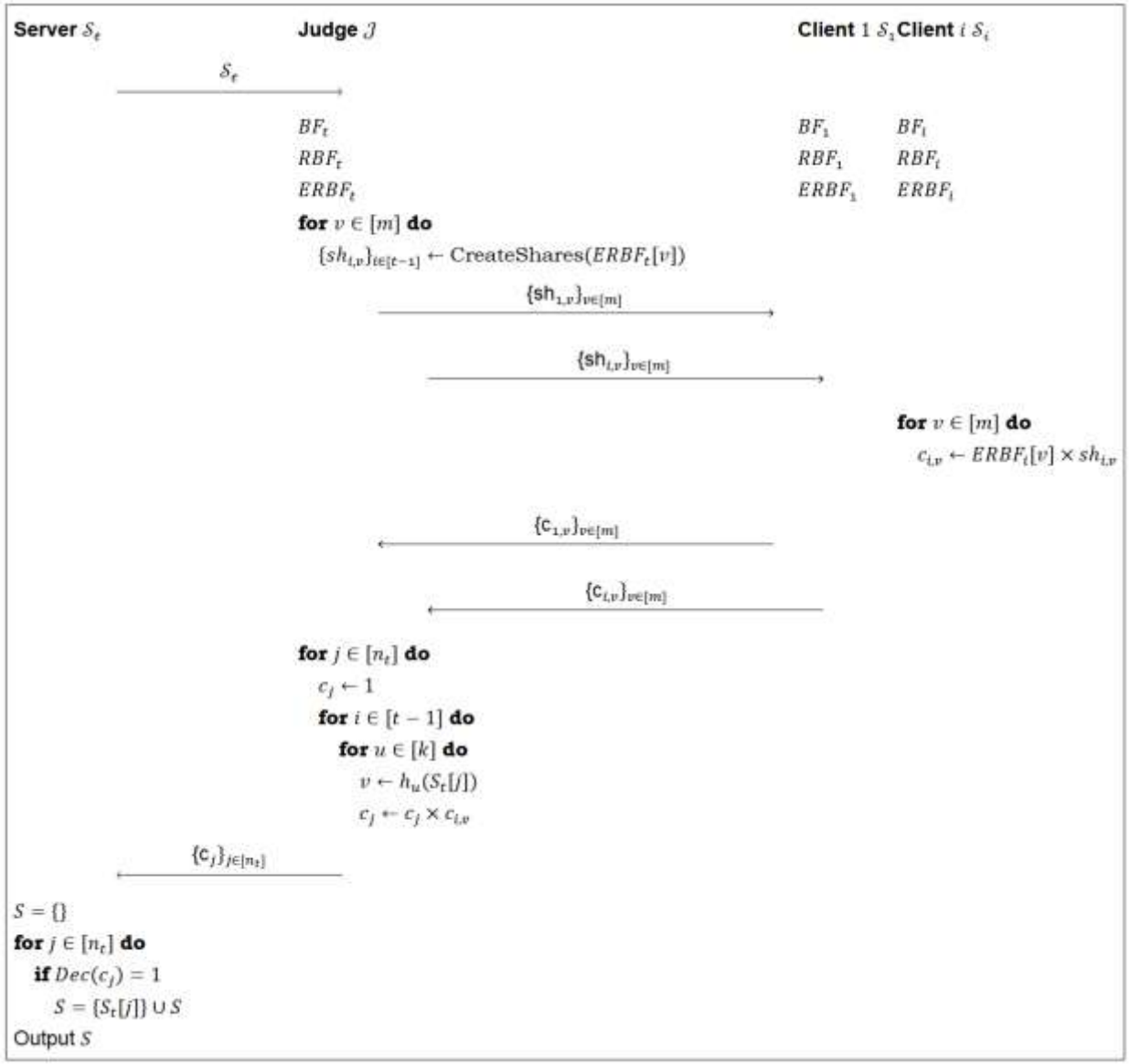
$\boldsymbol{ERBF_i}$ = the encrypted randomized Bloom filter of party i



Progress

APSI 2 – ~~InterActive~~ Judge

- The Judge receives the server's elements and computes an ERBF with them, then creates shares for each bin of the ERBF
- The Judge sends each client their shares, to which the clients add, for each bin, the bin of their own ERBF, send this ciphertext back to the Judge
- For each element of the server, the Judge aggregates all bins that the element hashes to from all ciphertexts, sends these aggregations to the server
- The server decrypts, adds the element to the intersection if the decryption is 1



Progress

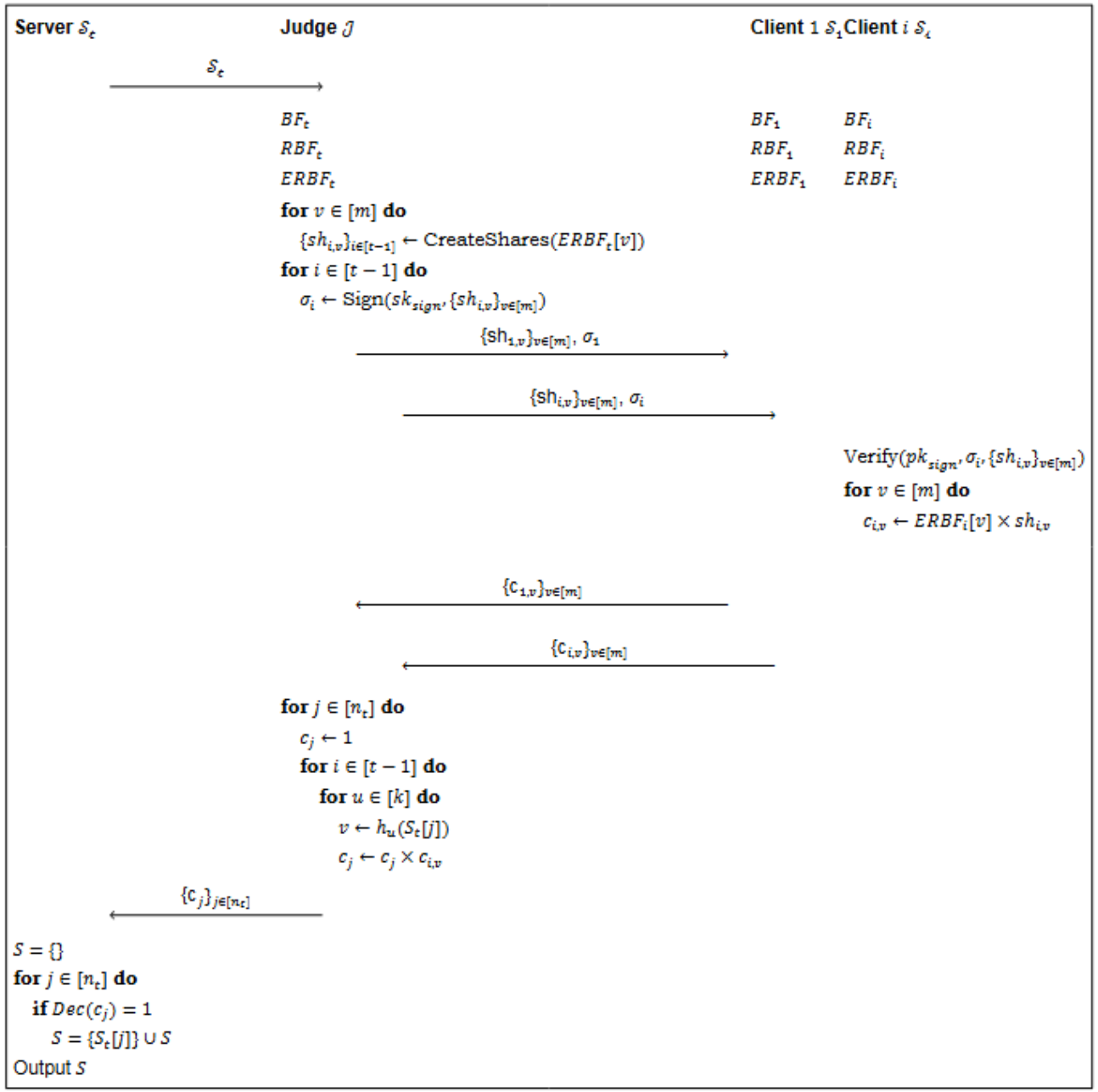
APSI 2 – ~~Inter~~Active Judge

Threat Model:

- Malicious Server
- (Semi?-)Honest Judge
- Semi-honest clients

\+ Server doesn't have to know the hash functions

\- Judge has to wait on computation from clients



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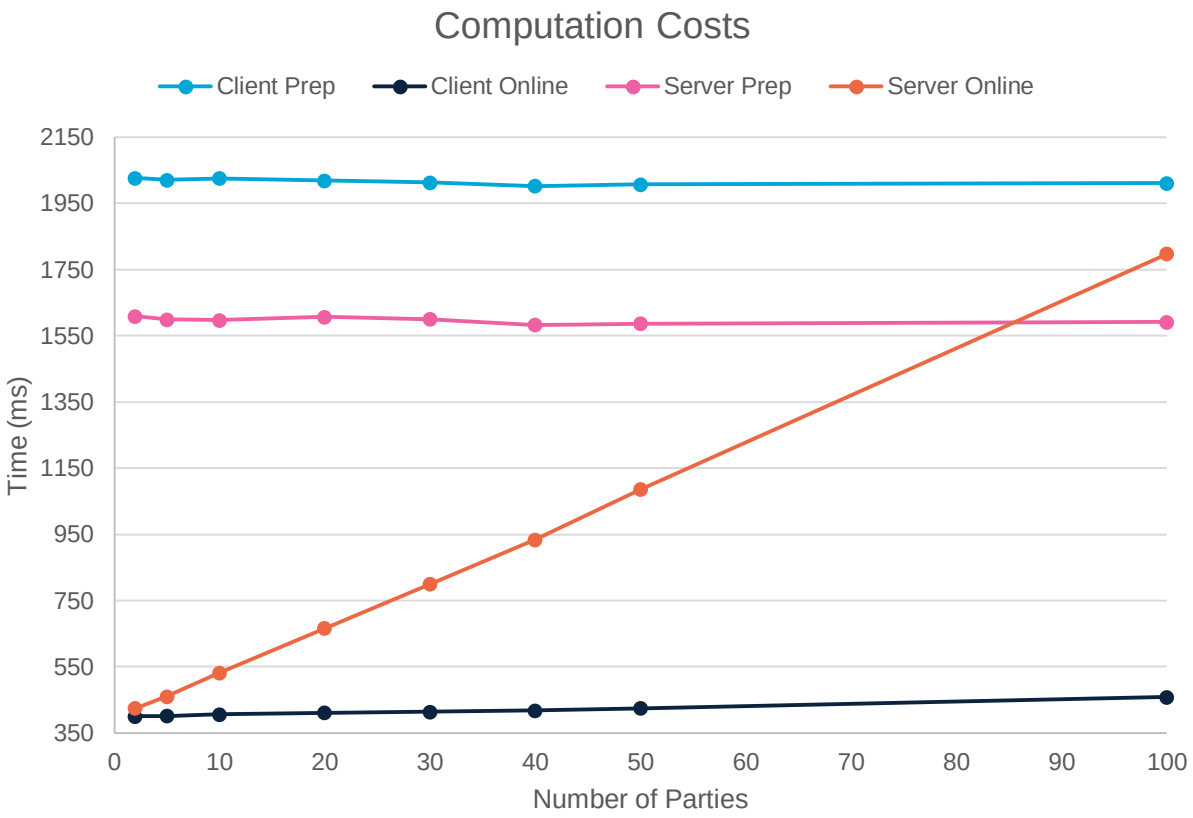
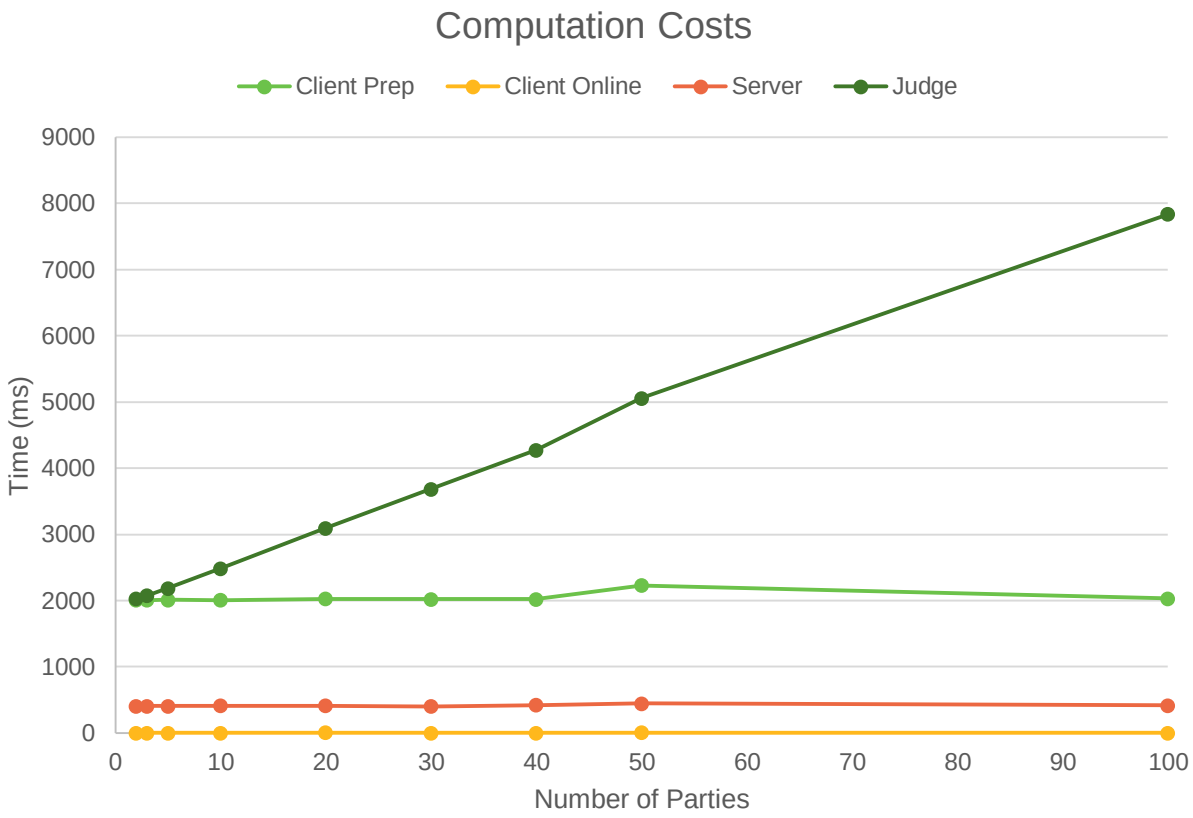
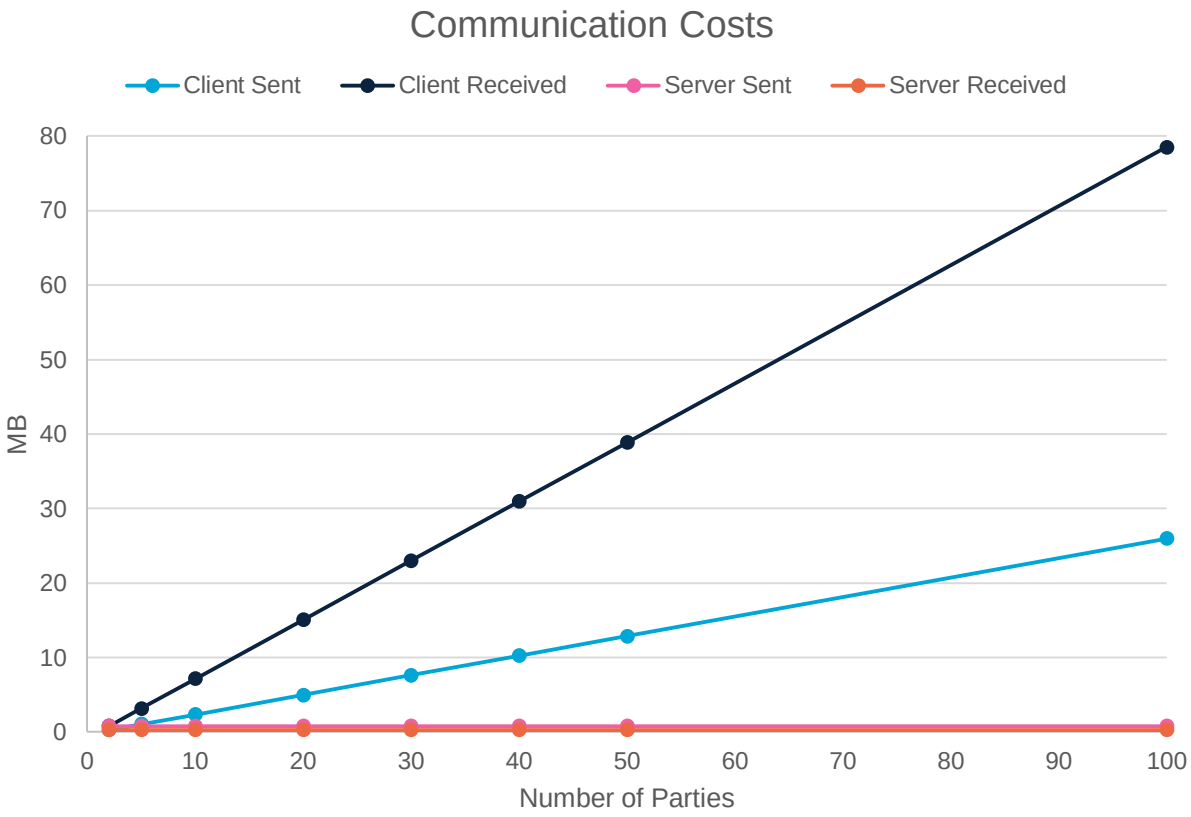


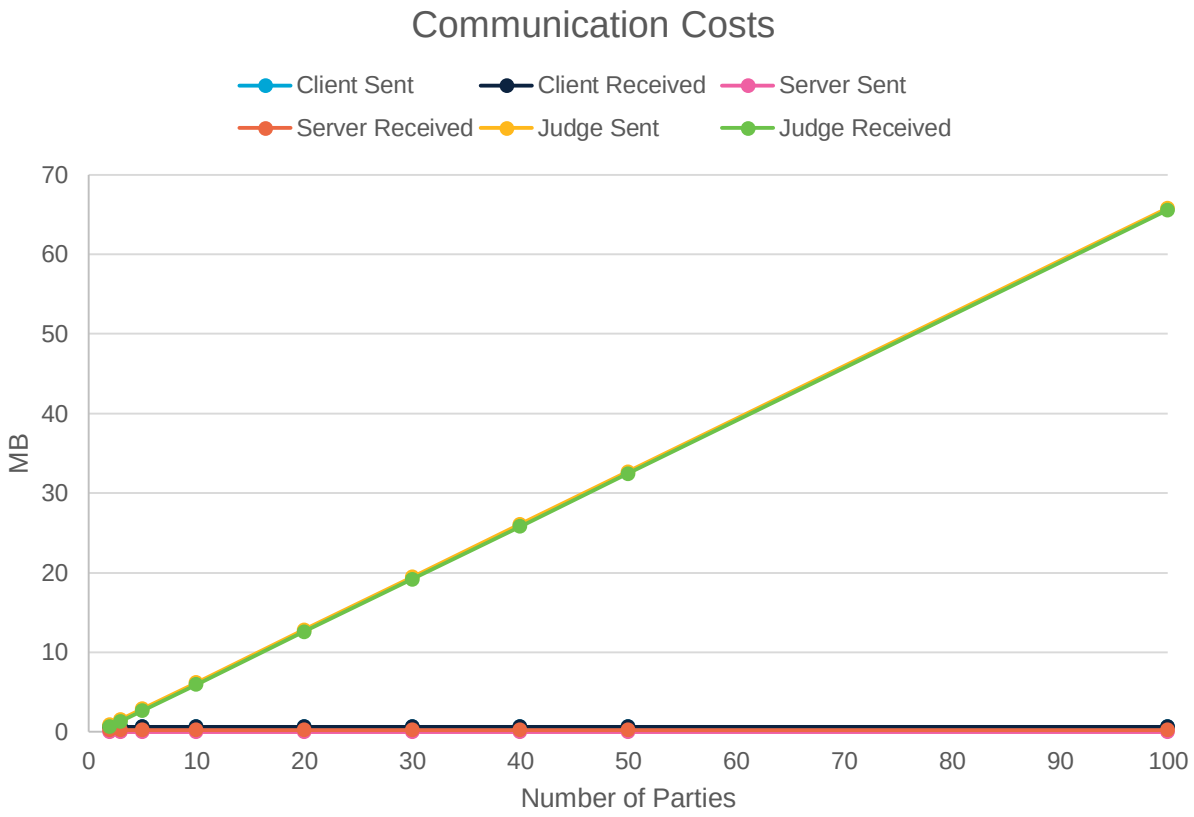
APSI2



Baseline



APSI2



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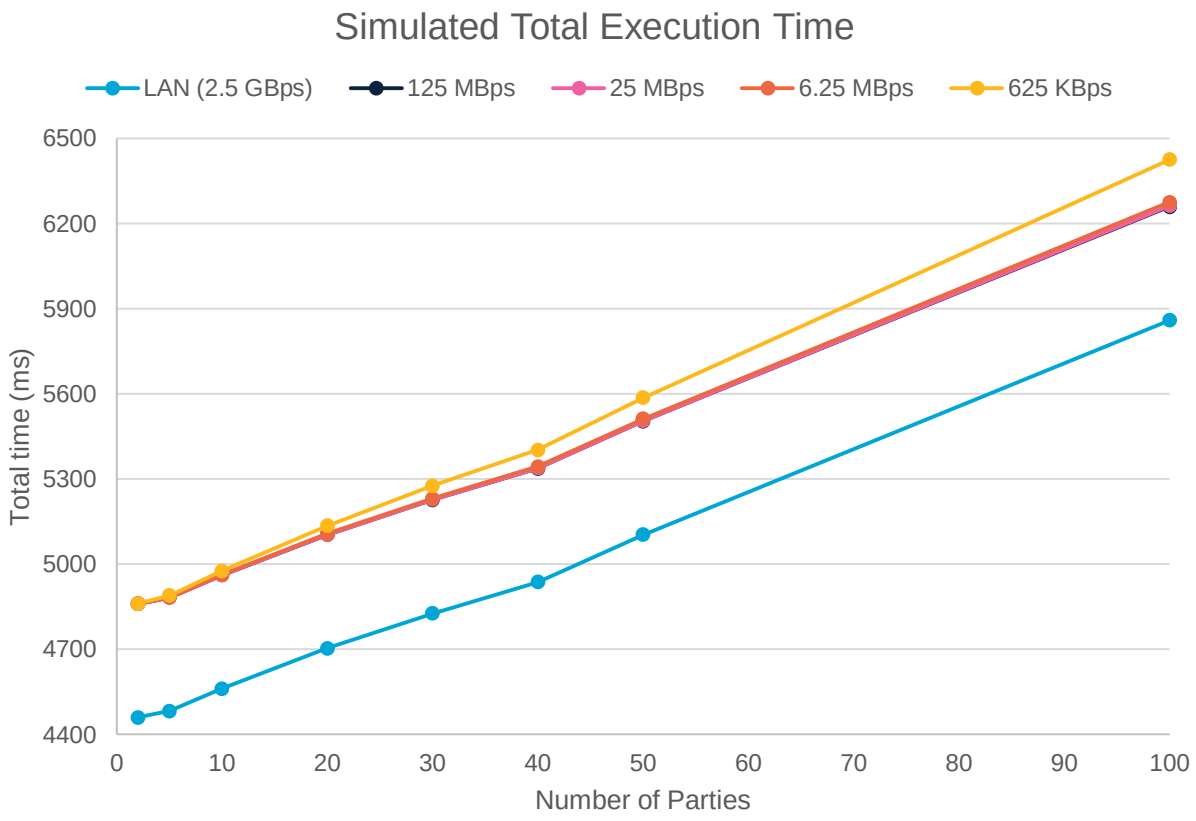
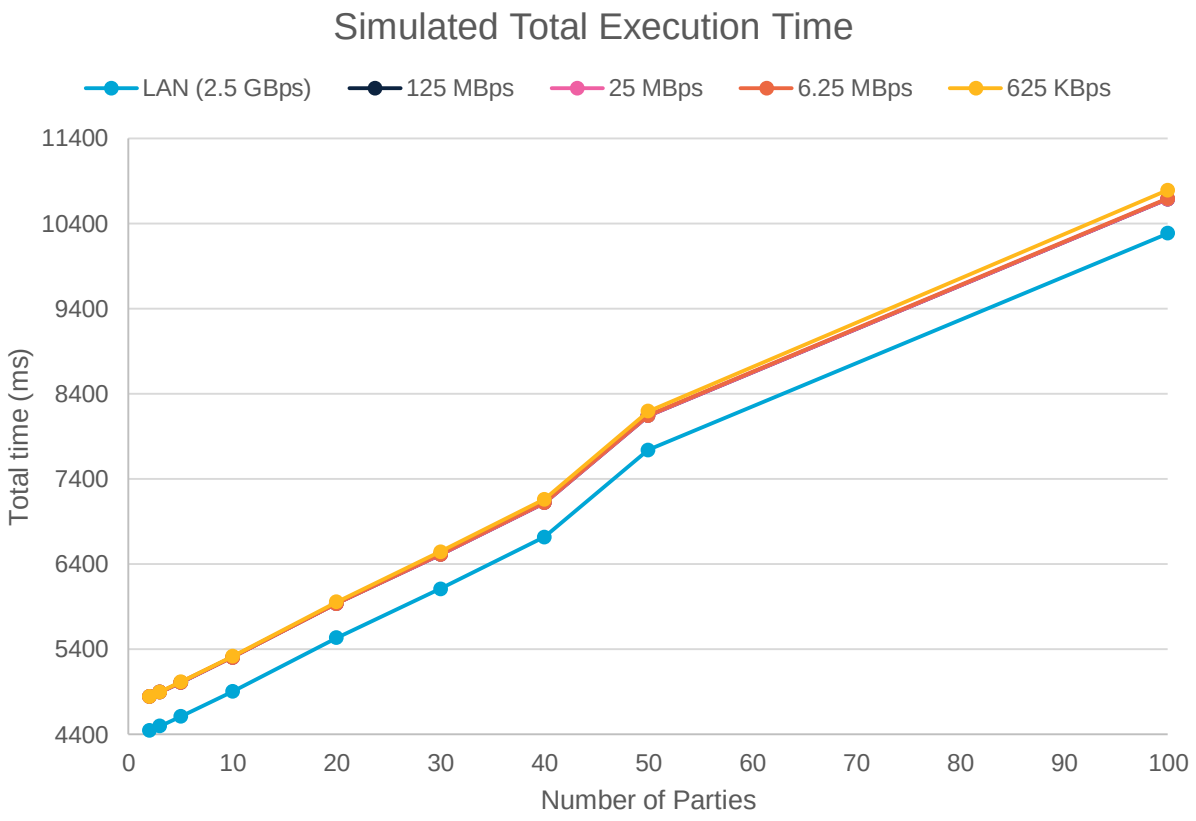


APSI2



Next Week

- Implement PAPSI2
  - Look more closely and try to understand the code of [4] (Lilika's APSI & PAPSI) and [5] (Tjitske's PAPSI)
- Brainstorm on APSI1
- Start thinking about the proof for APSI2
  - Finish reading [6] ("How to simulate it–a tutorial on the simulation proof technique")

Thank you!

19-03-2026